

Mathematics for AI

Key Concepts • Formulas • Examples • Applications

1. Why Math for AI?

- Helps us understand how AI models work.
- Useful for building, improving and debugging models.
- Foundations for machine learning, deep learning and optimization.
- Builds problem-solving and analytical skills.
- Helps us move from using AI to creating AI.

“Math turns ideas into intelligent systems.”

2. Linear Algebra

- Works with vectors, matrices and their operations.
- Used in almost every AI model (neural networks, embeddings, etc.).

Matrix Multiplication:

$$C = A \times B$$

If A is $m \times n$ and B is $n \times p$ then C is $m \times p$.

Example:

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \quad B = \begin{bmatrix} 2 & 0 \\ 1 & 2 \end{bmatrix}$$
$$A \times B = \begin{bmatrix} 4 & 4 \\ 10 & 8 \end{bmatrix}$$

3. Calculus

- Helps us understand change.
- Used in optimization (e.g., training AI models).

Derivative:

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

- Shows how a function changes at a point.
- Used to find gradients in machine learning.

Example:

$$f(x) = x^2$$
$$f'(x) = 2x$$

4. Probability & Statistics

- Helps us deal with uncertainty and make predictions.
- Used in data analysis, uncertainty estimation, model training, and evaluation.

Bayes' Theorem:

$$P(A | B) = \frac{P(B | A) P(A)}{P(B)}$$

- $P(A | B)$: probability of A given B .
- Used in spam detection, medical diagnosis, recommendation systems, etc.

Example:

$$P(\text{Spam} | \text{Word}) = \frac{P(\text{Word} | \text{Spam}) P(\text{Spam})}{P(\text{Word})}$$

5. Optimization

- Used to find the best values (e.g., model parameters).
- Optimization algorithms minimize or otherwise optimize an objective function during training.

Gradient Descent:

$$\theta = \theta - \alpha \nabla J(\theta)$$

- θ : model parameters
- α : learning rate
- $\nabla J(\theta)$: gradient of the loss function

Example:

$$\theta = 0.5, \quad \alpha = 0.1, \quad \nabla J(\theta) = 2$$
$$\theta_{\text{new}} = 0.5 - 0.1 \times 2 = 0.3$$

6. Vector Calculus

- Deals with vectors and multivariable functions.
- Used in multivariable optimization and deep learning.

Gradient (Multivariable):

$$\nabla f(x, y) = \left[\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right]$$

Example:

$$f(x, y) = x^2 + y^2$$
$$\nabla f(x, y) = [2x, 2y]$$
$$\nabla f(1, 1) = [2, 2]$$

7. Eigenvalues & Eigenvectors

- Help identify important directions and transformations in data.
- Used in PCA, dimensionality reduction, and deep learning.

Eigenvalue Equation:

$$A v = \lambda v$$

- v : eigenvector (non-zero vector)
- λ : eigenvalue (scalar)

Example:

$$A = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$$

Eigenvalues: $\lambda_1 = 2, \lambda_2 = 3$
Eigenvectors: $v_1 = [1, 0], v_2 = [0, 1]$

8. Common Distributions

- Used to model real-world data and uncertainty.
- Important in machine learning and Bayesian methods.

Distribution	PDF / PMF	Used For
Normal (Gaussian)	$\frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$	Real-valued data
Bernoulli	$P(X=1) = p$	Binary outcomes (0 or 1)
Binomial	$P(X=k) = \binom{n}{k} p^k (1-p)^{n-k}$	Number of successes
Poisson	$P(X=k) = \frac{e^{-\lambda} \lambda^k}{k!}$	Count data events

9. Useful Math Concepts

- ✓ Vectors and matrices
- ✓ Derivatives and gradients
- ✓ Probability and distributions
- ✓ Expectation and variance
- ✓ Optimization (gradient descent)
- ✓ Eigenvalues and PCA
- ✓ Linear transformations
- ✓ Multivariable calculus

10. Real-World Applications



Neural Networks

Linear algebra + calculus + optimization



Recommendation Systems

Probability + statistics



Computer Vision

Linear algebra + calculus + optimization



Natural Language Processing

11. Learning Path

- 1 Brush up on high-school mathematics.
- 2 Learn linear algebra.
- 3 Study calculus (single & multivariable).
- 4 Understand probability and statistics.
- 5 Learn optimization techniques.
- 6 Explore advanced topics as needed.
- 7 Practice with real AI problems.

12. Key Takeaways

- ✓ Math is the foundation of AI.
- ✓ Focus on understanding, not just memorizing formulas.
- ✓ Practice with examples and real datasets.
- ✓ Use Python libraries (NumPy, SciPy) to apply math concepts.
- ✓ The more you practice, the more

Linear Algebra Cheat Sheet

 = Matrix / Vector
-- = Operation / Action

The essential concepts and formulas you need to know!

1. Scalars, Vectors, Matrices Concept

Scalar
(1x1)

5

Vector
(n x 1)

$\begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$

Matrix
(m x n)

$\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}$

- Scalar is a single number
- Vector has magnitude and direction
- Matrix is a 2D array of numbers

2. Matrix Addition & Subtraction Formula

$$A + B = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} + \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix} = \begin{bmatrix} a_{11} + b_{11} & a_{12} + b_{12} \\ a_{21} + b_{21} & a_{22} + b_{22} \end{bmatrix}$$

- Matrices must have the same dimensions
- Operation is element-wise

3. Matrix Multiplication Formula

A (m x n)
 $\begin{bmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{m1} & \cdots & a_{mn} \end{bmatrix}$

$\times$

B (n x p)
 $\begin{bmatrix} b_{11} & \cdots & b_{1p} \\ \vdots & \ddots & \vdots \\ b_{n1} & \cdots & b_{np} \end{bmatrix}$

$=$

C (m x p)
 $\begin{bmatrix} c_{11} & \cdots & c_{1p} \\ \vdots & \ddots & \vdots \\ c_{m1} & \cdots & c_{mp} \end{bmatrix}$

- $(AB)_{ij} = \sum_{k=1}^n a_{ik} b_{kj}$
- Columns of A must match rows of B

4. Transpose Formula

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \xrightarrow{\text{Transpose}} A^T = \begin{bmatrix} a_{11} & a_{21} & a_{31} \\ a_{12} & a_{22} & a_{32} \\ a_{13} & a_{23} & a_{33} \end{bmatrix}$$

- Rows become columns
- $(A^T)^T = A$

5. Determinant (2x2, 3x3) Formula

2x2 Matrix
 $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$

$|A| = ad - bc$

3x3 Matrix
 $\begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$

$|A| = aei + bfg + cdh - ceg - bdi - afh$

6. Inverse Matrix Formula

For an invertible matrix A

$$A A^{-1} = A^{-1} A = I$$

2x2 Matrix
 $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$

$\rightarrow A^{-1} = \frac{1}{ad-bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$

(where $ad - bc \neq 0$)

7. Identity Matrix Concept

$$I_n = \begin{bmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{bmatrix}$$

- Ones on the main diagonal
- Zeros elsewhere
- $AI = IA = A$

8. Dot Product Formula

$$x = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \quad y = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix} \quad x \cdot y = \sum_{i=1}^n x_i y_i$$

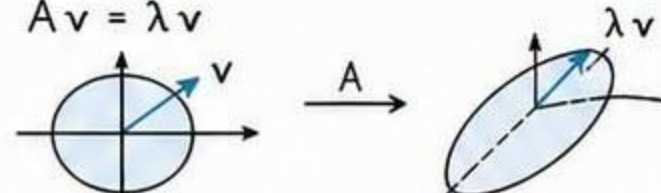
- Result is a scalar
- Measures similarity

9. Norm (Length) Formula

$$x = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \quad \|x\|_2 = \sqrt{\sum_{i=1}^n x_i^2}$$

- Euclidean (L2) norm
- $\|x\|_2$ is always ≥ 0

10. Eigenvalues & Eigenvectors Concept

$$Av = \lambda v$$


- v is eigenvector
- λ is eigenvalue
- $Av = \lambda v$ (same or opposite direction depending on λ)

11. Rank of a Matrix Concept

$$\text{rank}(A) = \begin{array}{|l} \text{Maximum number of} \\ \text{linearly independent} \\ \text{rows or columns} \end{array}$$

- $\text{rank}(A) \leq \min(m, n)$
- Full rank if $\text{rank}(A) = \min(m, n)$

12. System of Linear Equations Concept

$$Ax = b$$
$$A = \begin{bmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{m1} & \cdots & a_{mn} \end{bmatrix} \quad x = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} \quad b = \begin{bmatrix} b_1 \\ \vdots \\ b_m \end{bmatrix}$$

- Solve for x
- Unique solution if A is square and full rank
- No solution or infinite solutions otherwise

Quick Tips

- Check matrix dimensions
- Understand operations before applying
- Practice with small examples

Tools to Use

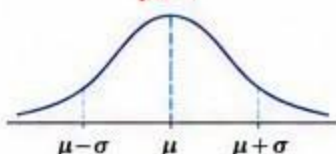
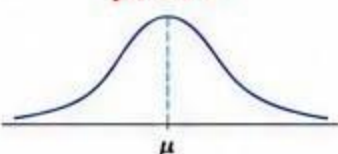
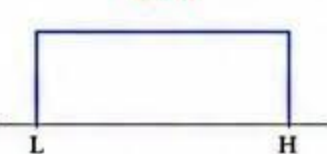
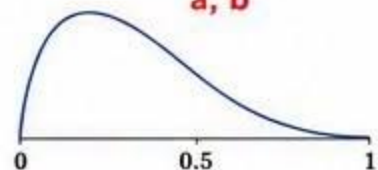
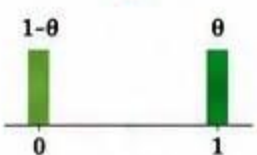
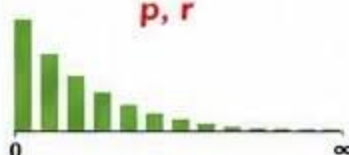
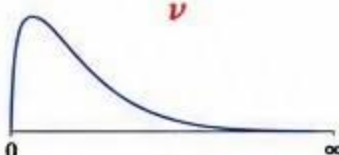
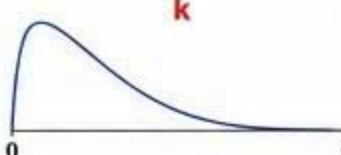
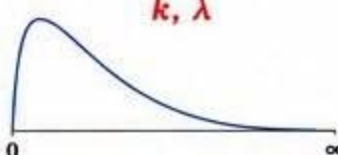
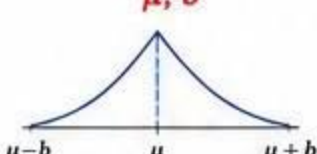
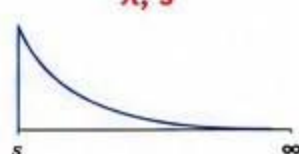
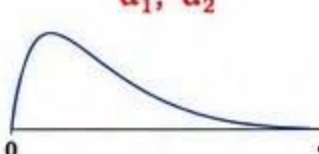
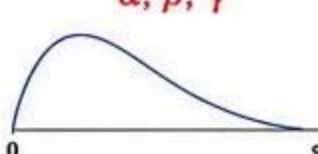
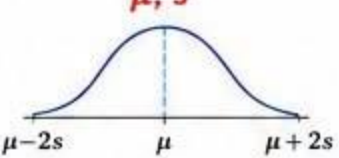
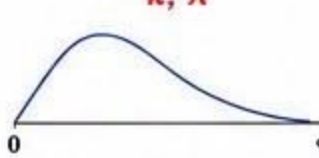
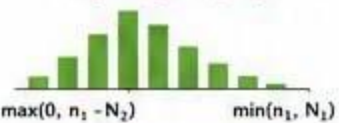
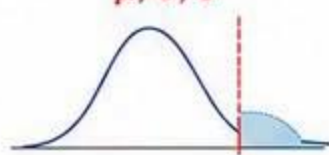
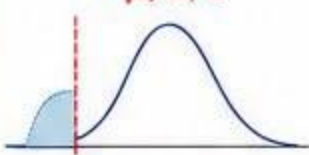
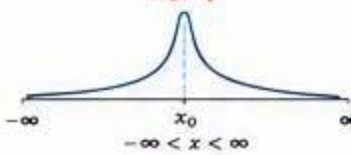
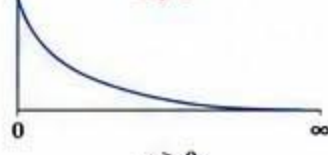
- NumPy (Python)
- MATLAB / Octave
- Jupyter Notebook
-

Why It Matters

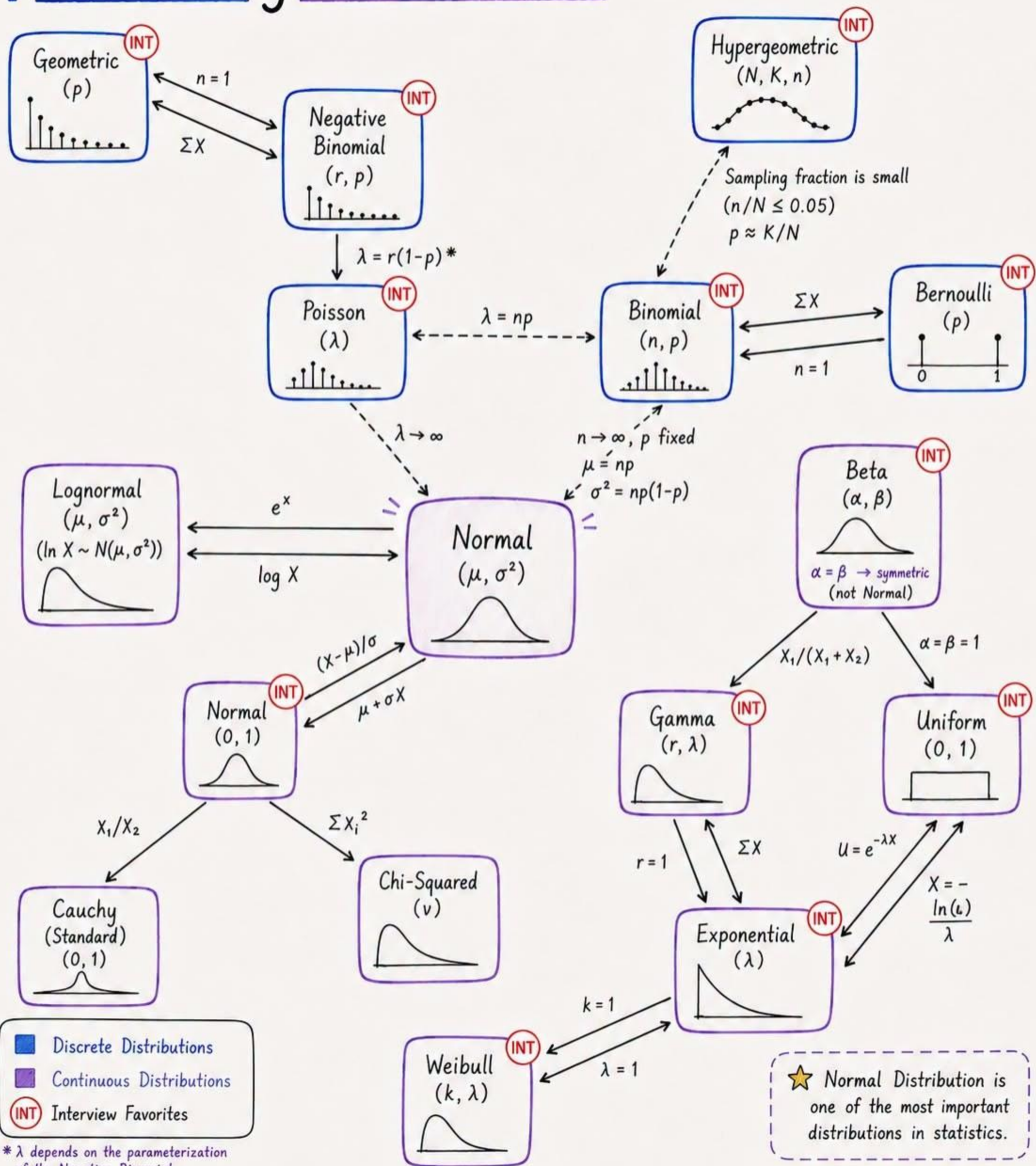
- Foundation for ML & AI
- Used in transformations
- Powers algorithms like PCA, SVD, Neural Networks

Probability Distributions

Continuous (PDF) – defined over real values Discrete (PMF) – defined over integers

Normal μ, σ  $-\infty < x < \infty$ $f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$ Symmetric bell-shaped distribution.	t-distribution μ, σ, ν  $-\infty < x < \infty$ $f(x) = \frac{\Gamma\left(\frac{\nu+1}{2}\right)}{\sigma\sqrt{\nu\pi}\Gamma\left(\frac{\nu}{2}\right)} \left(1 + \frac{\left(\frac{x-\mu}{\sigma}\right)^2}{\nu}\right)^{-\frac{\nu+1}{2}}$ Like normal but heavier tails; used for small samples.	Uniform L, H  $L \leq x \leq H$ $f(x) = \frac{1}{H-L}$ All outcomes in $[L, H]$ are equally likely.	Beta a, b  $0 < x < 1$ $f(x) = \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)} x^{a-1}(1-x)^{b-1}$ Defined on $(0, 1)$; flexible for shapes.	
Beta-Binomial a, b, n  $k = 0, 1, \dots, n$ $P(X = k) = \binom{n}{k} \frac{B(k+a, n-k+b)}{B(a, b)}$ Binomial with random success probability.	Bernoulli θ  $k = 0, 1$ $P(X = x) = \theta^x(1-\theta)^{1-x}$ Single trial with two outcomes.	Gamma α, β  $x > 0$ $f(x) = \frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\beta x}$ Positive continuous data; waiting times, lifetimes.	Inverse-Gamma α, β  $x > 0$ $f(x) = \frac{\beta^\alpha}{\Gamma(\alpha)} x^{-\alpha-1} e^{-\frac{\beta}{x}}$ Modeling variances and scale parameters.	Binomial p, n  $k = 0, 1, \dots, n$ $P(X = k) = \binom{n}{k} p^k(1-p)^{n-k}$ Number of successes in n trials.
Negative Binomial p, r  $k = 0, 1, 2, \dots$ $P(X = k) = \binom{k+r-1}{k} p^r(1-p)^k$ Number of failures before r successes.	Folded t ν  $x \geq 0$ Distribution of $ T $ where T follows t-distribution.	Poisson λ  $k = 0, 1, 2, \dots$ $P(X = k) = e^{-\lambda} \frac{\lambda^k}{k!}$ Count of events in a fixed interval.	Chi-square k  $x > 0$ $f(x) = \frac{1}{2^{k/2}\Gamma(k/2)} x^{\frac{k}{2}-1} e^{-\frac{x}{2}}$ Sum of squares of k standard normal variables.	Noncentral Chi-square k, λ  $x > 0$ Generalization of chi-square; used in power analysis.
Laplace (Double Exponential) μ, b  $-\infty < x < \infty$ $f(x) = \frac{1}{2b} e^{-\frac{ x-\mu }{b}}$ Symmetric, sharper peak and heavier tails.	Exponential λ  $x \geq 0$ $f(x) = \lambda e^{-\lambda x}$ Time between events in a Poisson process.	Shifted Exponential λ, s  $x \geq s$ $f(x) = \lambda e^{-\lambda(x-s)}$ Exponential with a minimum starting value s .	F-distribution d_1, d_2  $x > 0$ Ratio of two scaled chi-square variables.	Generalized Gamma α, β, γ  $x > 0$ General family including gamma, Weibull, and exponential.
Logistic μ, s  $-\infty < x < \infty$ $f(x) = \frac{e^{-\frac{x-\mu}{s}}}{s \left(1 + e^{-\frac{x-\mu}{s}}\right)^2}$ Similar to normal, heavier tails.	Log-Normal μ, σ  $x > 0$ $f(x) = \frac{1}{x\sigma\sqrt{2\pi}} e^{-\frac{(\ln x - \mu)^2}{2\sigma^2}}$ Model for multiplicative processes.	Pareto x_m, α  $x \geq x_m$ $f(x) = \frac{\alpha x_m^\alpha}{x^{\alpha+1}}$ Heavy-tailed; 80/20 rule phenomena.	Weibull k, λ  $x \geq 0$ $f(x) = \frac{k}{\lambda} \left(\frac{x}{\lambda}\right)^{k-1} e^{-\left(\frac{x}{\lambda}\right)^k}$ Lifetimes, reliability analysis.	Categorical $p_1, \dots, p_k$  $\sum_{i=1}^k p_i = 1$ $P(X = i) = p_i$ Multinomial with one trial; more than two categories.
Wallenius' Noncentral Hypergeometric N_1, N_2, n_1, ψ  k in valid range Sampling without replacement with unequal odds (Wallenius').	Normal (Right-Censored) μ, σ, c  $-\infty < x < \infty$ Values above c are censored.	Normal (Left-Censored) μ, σ, c  $-\infty < x < \infty$ Values below c are censored.	Cauchy x_0, γ  $-\infty < x < \infty$ $f(x) = \frac{1}{\pi\gamma \left[1 + \left(\frac{x-x_0}{\gamma}\right)^2\right]}$ Heavy tails; mean and variance undefined.	Half-t σ, ν  $x \geq 0$ Positive half (absolute value) of a Student's t-distribution.

Probability Distributions



CENTRAL LIMIT THEOREM (CLT)

A KEY CONCEPT IN STATISTICS

The Central Limit Theorem (CLT) states that the sampling distribution of the mean becomes approximately normal as the sample size increases, provided the observations are independent and the population has a finite mean (μ) and finite variance (σ^2).

1. DEFINITION

If we take many random samples of size n from a population (with finite mean μ and finite variance σ^2), then the distribution of the sample means ($\bar{x}$) will be approximately normal as n becomes large.

This is true even if the original population is not normally distributed, as long as it has a finite mean and variance and the samples are independent.

2. CENTRAL LIMIT THEOREM FORMULA

The sampling distribution of the mean follows an approximately normal distribution:

$$\bar{x} \sim N\left(\mu, \frac{\sigma^2}{n}\right)$$

Where:

$\bar{x}$ = sample mean
 μ = population mean
 σ^2 = population variance
 n = sample size

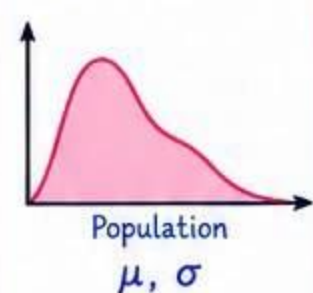
The standard deviation of this distribution (i.e., Standard Error) is $\sigma_{\sqrt{n}}$.

3. CONDITIONS FOR CLT

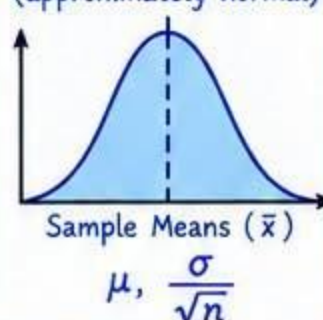
- 1 Random Sampling**
The samples should be randomly selected and independent.
- 2 Large Sample Size**
Generally, $n \geq 30$ is considered large (some cases work with smaller n).
- 3 Finite Mean and Variance**
The population must have a finite mean (μ) and finite variance (σ^2).
- 4 Independence**
Observations in the sample should be independent of each other.

4. POPULATION vs. SAMPLING DISTRIBUTION

Population Distribution
(may be non-normal)



Sampling Distribution
of the Mean
(approximately normal)



5. SAMPLING DISTRIBUTION OF THE MEAN

If we draw many samples of size n and calculate the mean of each sample, the distribution of those sample means is approximately normal with:

$$\begin{aligned}\text{Mean} &= \mu \\ \text{Standard Deviation (Standard Error)} &= \frac{\sigma}{\sqrt{n}}\end{aligned}$$

6. WHY SAMPLE SIZE MATTERS?

As the sample size (n) increases:

- The sampling distribution becomes more normal.
- The standard error $\left(\frac{\sigma}{\sqrt{n}}\right)$ decreases.
- The sample mean generally becomes a more precise estimate of the population mean.

Rule of Thumb: $n \geq 30$ is commonly used, but the required sample size depends on the population's distribution.

7. STANDARD ERROR FORMULA

The standard deviation of the sampling distribution of the mean is called the Standard Error (SE).

$$SE = \frac{\sigma}{\sqrt{n}}$$

Where:

σ = population standard deviation
 n = sample size

A smaller SE means sample means tend to vary less around the population mean.

8. EXAMPLE

Suppose the heights in a population have a mean (μ) = 170 cm and standard deviation (σ) = 10 cm. We take random samples of size $n = 36$.

$$SE = \frac{\sigma}{\sqrt{n}} = \frac{10}{\sqrt{36}} = \frac{10}{6} = 1.67$$

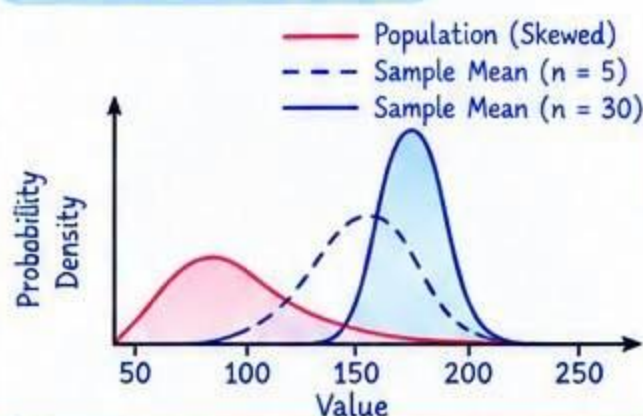
According to the CLT:

$$\bar{x} \sim N\left(170, \frac{10^2}{36}\right)$$

Interpretation:

The distribution of sample means (for $n = 36$) will be approximately normal with mean 170 and standard deviation 1.67.

9. VISUAL EXAMPLE



As the sample size increases, the sampling distribution of the mean becomes more normal and less spread out.

10. APPLICATIONS IN DATA SCIENCE

- ✓ Hypothesis Testing
- ✓ Confidence Intervals
- ✓ A/B Testing
- ✓ Estimating Population Parameters
- ✓ Machine Learning (e.g., model evaluation)
- ✓ Data Analysis and Decision Making
- ✓ Quality Control
- ✓ Survey Analysis

11. ADVANTAGES

- ✓ Works for many non-normal population distributions, given appropriate conditions.
- ✓ Simple and powerful concept.
- ✓ Allows inference about population using sample data.
- ✓ Forms the basis of many statistical methods.
- ✓ Widely used in real-world data analysis.

12. LIMITATIONS

- ✗ May not work well with small sample sizes ($n < 30$) for highly non-normal distributions.
- ✗ Requires finite mean and variance in the population.
- ✗ For highly skewed populations or populations with extreme outliers, a much larger sample size may be required.

13. KEY POINTS TO REMEMBER

- ★ The CLT applies to the sampling distribution of the mean.
- ★ The sampling distribution is approximately normal for large n .
- ★ Mean of the sampling distribution = population mean (μ).
- ★ Standard deviation of the sampling distribution = $\frac{\sigma}{\sqrt{n}}$ (Standard Error).
- ★ $n \geq 30$ is a common rule of thumb, but it depends on the population.
- ★ The CLT is the foundation for hypothesis testing and confidence intervals.

Hypothesis Testing

Concepts • Tests • Examples • Python

1. What is Hypothesis Testing?

A statistical method to decide whether there is enough evidence to reject a claim about a population.

- Helps us make decisions using data.
- Commonly used in business, science, and data analysis.
- Based on sample data, not the entire population.

Example:

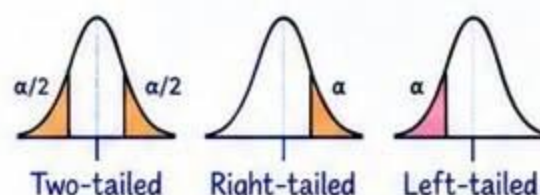
Is a new teaching method better than the old one?

2. Key Terminology

- Null Hypothesis (H_0):** Default claim (no effect or difference).
- Alternative Hypothesis (H_1):** What we want to prove (effect or difference).
- Test statistic:** A value calculated from sample data.
- p-value:** Probability of observing results at least as extreme as the sample result, assuming H_0 is true.
- Significance Level (α):** Threshold for decision (commonly 0.05).
- Reject H_0 :** If p-value $\leq \alpha$.
- Fail to reject H_0 :** If p-value $> \alpha$.

3. Types of Hypothesis Tests

- Two-tailed test:** Checks for any difference ($\neq$).
- Right-tailed test:** Checks if the value is greater ($>$).
- Left-tailed test:** Checks if the value is smaller ($<$).



4. Steps in Hypothesis Testing

- State H_0 and H_1
- Choose the significance level (α)
- Select the appropriate test
- Calculate the test statistic and p-value
- Make a decision (reject or fail to reject H_0)
- Interpret the result in context

Common α values:

0.10 (10%), 0.05 (5%), 0.01 (1%)

5. Common Hypothesis Tests

Test	Used For	Example
Z-test	Population mean (σ known)	Mean, proportion
t-test	Population mean (σ unknown)	Mean
Chi-square test	Categorical data	Goodness of fit, independence
ANOVA	Compare 3+ group means	More than two groups
F-test	Compare two population variances	Two sample variances

6. Z-test (Population Mean)

Used when population standard deviation (σ) is known. Large samples are often preferred for normal approximation.

$$z = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}}$$

- $\bar{x}$ = sample mean
- μ = population mean (under H_0)
- σ = population standard deviation
- n = sample size

Example:

$H_0: \mu = 50$
 $H_1: \mu \neq 50$

7. t-test (Population Mean)

Used when population standard deviation (σ) is unknown, especially for small samples.

$$t = \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}}$$

- $\bar{x}$ = sample mean
- μ = population mean (under H_0)
- s = sample standard deviation
- n = sample size
- Degrees of freedom = $n - 1$

Example:

$H_0: \mu = 100$
 $H_1: \mu > 100$ (right-tailed)

8. Chi-Square Test

Used for categorical data to check goodness of fit or independence.

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

- O = observed frequency
- E = expected frequency
- Degrees of freedom = $(\text{rows} - 1) \times (\text{columns} - 1)$

Example:

H_0 : Variables are independent
 H_1 : Variables are not independent

9. ANOVA (F-test)

Used to compare the means of three or more groups.

$$F = \frac{\text{Between-group variance}}{\text{Within-group variance}}$$

- H_0 : All group means are equal.
- H_1 : At least one group mean is different.
- If p-value $\leq \alpha$, reject H_0 .

Example:

10. Python Example (t-test)

```
import numpy as np
from scipy import stats

# Sample data
sample = [98, 102, 105, 100, 96]

# One-sample t-test
# H0: mean = 100
result = stats.ttest_1samp(sample, popmean=100)

print("t-statistic:", result.statistic)
print("p-value:", result.pvalue)
```

Example Output:

11. Python Example (Chi-Square)

```
import numpy as np
from scipy.stats import chi2_contingency

# Observed frequencies
observed = np.array([[20, 30],
                     [15, 35]])

chi2, p, dof, expected = chi2_contingency(
    observed)

print("Chi2 statistic:", chi2)
print("p-value:", p)
print("Degrees of freedom:", dof)
```

Example Output:

12. Important Tips

- ✓ Check assumptions before choosing a test.
- ✓ Use the right test for your data type and sample size.
- ✓ A small p-value ($\leq \alpha$) means strong evidence against H_0 .
- ✓ Statistical significance $\neq$ practical significance.
- ✓ Always interpret results in the context of the problem.
- ✓ Visualize your data (boxplots, histograms) before testing.

STANDARD NORMAL DISTRIBUTION (Z-DISTRIBUTION)

The Standard Normal Distribution is a special type of normal distribution that has Mean (μ) = 0 and Standard Deviation (σ) = 1.

1. DEFINITION

A normal distribution with:

- Mean (μ) = 0
- Standard Deviation (σ) = 1

It is also called the Z-Distribution because every value is converted into a Z-score.

2. FORMULA (PDF)

The probability density function is:

$$f(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}}$$

Where:

- z = Standard score (Z-score)
- e = Euler's number (≈ 2.718)
- π = 3.14159

3. Z-SCORE FORMULA

A value from any normal distribution can be converted into a Z-score.

$$z = \frac{x - \mu}{\sigma}$$

Where:

- x = Observed value
- μ = Population mean
- σ = Population standard deviation
- z = Standard score

4. CHARACTERISTICS

1 Bell-Shaped Curve

The graph forms a symmetric bell shape.



2 Mean = Median = Mode

All three are equal.

$$\text{Mean} = \text{Median} = \text{Mode} = 0$$

3 Symmetrical Distribution

Left and right sides are mirror images.



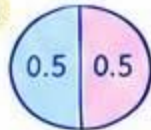
4 Total Area = 1 (100%)

The total probability under the curve equals 1.

1.000

5 Half Area on Each Side

Left side = 0.5
Right side = 0.5



6 Infinite Tails

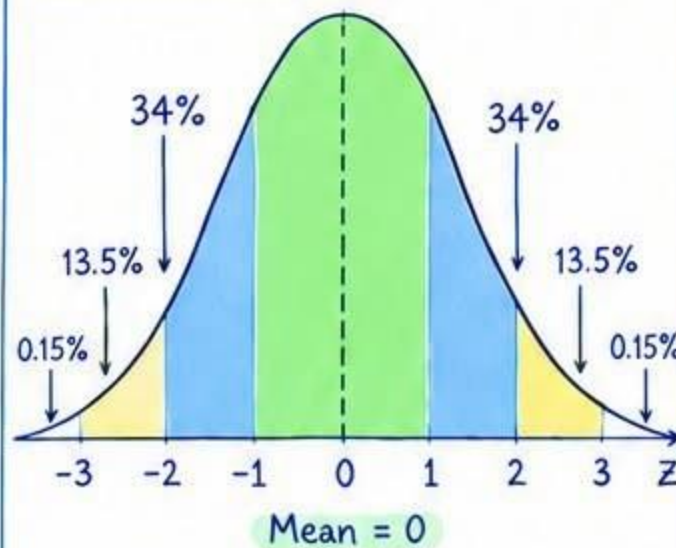
The curve extends indefinitely in both directions but never touches the x-axis.

7 Continuous Distribution

It is defined for every real number.

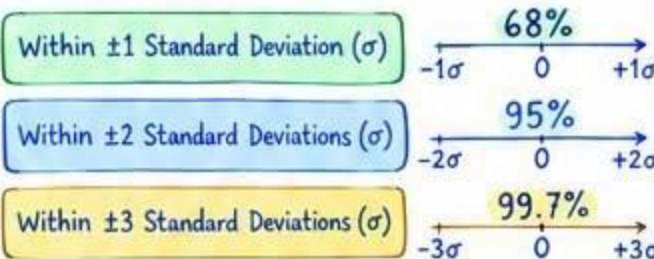
$$-\infty < Z < +\infty$$

5. SHAPE OF STANDARD NORMAL DISTRIBUTION



6. EMPIRICAL RULE (68-95-99.7 RULE)

This rule describes how data are distributed around the mean.



Example: If students' heights follow a normal distribution:

- About 68% lie within ± 1 SD of the mean.
- About 95% lie within ± 2 SD.
- About 99.7% lie within ± 3 SD.

7. STANDARD NORMAL TABLE (Z-TABLE)

The Z-table gives the probability (area to the left of a Z-score).

Z-score	Probability $P(Z \leq z)$
-3.00	0.0013
-2.58	0.0049
-2.00	0.0228
-1.50	0.0668
-1.00	0.1587
-0.50	0.3085
0.00	0.5000
0.50	0.6915
1.00	0.8413
1.50	0.9332
2.00	0.9772
2.58	0.9951
3.00	0.9987



Tip: The table usually gives area to the left of z .

8. INTERPRETATION OF Z-SCORE

$Z = 0$	Exactly at the mean
$Z > 0$	Above the mean
$Z < 0$	Below the mean
$Z \geq +2$	Two SDs above the mean
$Z \leq -2$	Two SDs below the mean

9. SOLVED EXAMPLE

Suppose:
Mean (μ) = 70
Standard Deviation (σ) = 10
Student's Marks (x) = 85
Find the Z-score.

$$z = \frac{x - \mu}{\sigma} = \frac{85 - 70}{10} = \frac{15}{10} = 1.5$$

Interpretation:
The student scored 1.5 standard deviations above the mean.

10. APPLICATIONS

- ✓ Hypothesis Testing
- ✓ Z-Test
- ✓ Confidence Intervals
- ✓ Statistical Inference
- ✓ Machine Learning
- ✓ Data Science
- ✓ Quality Control
- ✓ Psychology
- ✓ Medical Research
- ✓ Economics
- ✓ Finance
- ✓ Educational Testing
- ✓ SPSS & Statistical Software

11. ADVANTAGES

- Simplifies probability calculations.
- Allows comparison of values from different datasets.
- Forms the basis of many statistical tests.
- Makes hypothesis testing easier.
- Helps detect unusual observations (outliers).

12. LIMITATIONS

- Assumes data are normally distributed.
- Mean and standard deviation can be affected by extreme outliers.
- Not suitable for highly skewed data.
- Best suited for continuous quantitative data.

★ KEY POINTS TO REMEMBER

- Standard Normal Distribution is a normal distribution with mean 0 and SD 1.
- Every value is expressed as a Z-score.
- The curve is bell-shaped and perfectly symmetric.
- The total area under the curve is 1 (100%).
- It follows the 68-95-99.7 empirical rule.
- It is fundamental in hypothesis testing, confidence intervals, probability calculations and many statistical analyses.